- *GraphQL* APIs allow a digital twin to request exactly the data that it needs in a single query which is valuable when integrating with systems that expose rich data models when you only need a few attributes.
- *gRPC* APIs provide high-performance, type-safe communication particularly valuable for real time operations. It's binary protocol and code generation from schema definitions reduce integration complexity while improving performance over HTTP-based alternatives.
- *WebSocket* APIs enable bidirectional, real-time communication between systems. Unlike request-response patterns, WebSockets maintain persistent connections that support instant data updates and control commands.

TRY IT OUT: ACQUIRE SATELLITE IMAGERY FROM A REST API

The NASA visible infrared imaging radiometer suite (VIIRS) instrument is a satellite mounted sensor that collects imagery of the earth's surface for monitoring and investigating changes in surface vegetation, and water and land use, taking daily images. You can understand how this information would be useful to a digital twin that needs to represent geophysical properties of part of the earth. NASA publishes a RESTful API as part of its global imagery browse service (GIBS) service, that you can call to get an image of the earth. The sample code in listing 4.6 retrieves an image, taken by VIIRS, of the largest man-made lake in the world which allows tracking the impact of drought on this important reservoir.